BACHELOR THESIS

On the Equilibrium of Trapped Polarizable Nanoparticles in Optical Cavities

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Declaration of Authorship

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Abstract

Driven by recent advancements in high-finesse cavities and nanoparticle control, levitodynamics has become a key framework for investigating fundamental physics and developing novel sensors. We investigate the dynamics of trapped nanoparticles, coupled to an optical cavity. We develop a theoretical model of the system and derived the equilibrium conditions. Numerical calculations indicate the existence of configurations with multiple equilibrium positions. We study the system generally with N particles and analyze the 1 and 2 particle cases. For 1 particle, our analysis indicates that all equilibrium positions are stable, within the studied parameter regime. For 2 particles we found that the solutions lie on a manifold in configuration space.

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1. Introduction

Recent developments led to the emergence of the field of levitodynamics. This field is primarily concerned with understanding, controlling, and using levitated nanoparticles. It does so by using techniques developed in other research fields such as atomic physics and biophysics [1]. A large contributor was 2018 Nobel Prize winner Arthur Ashkin [2, 3], whose early work on optical tweezers is resurfacing in this field [4]. His research was continued and current experimental work in this field has further advanced the frontier [5]. The high finesse cavities have also developed in recent times [6, 7]. Levitodynamics has already been successful in the production of precise and novel sensors [8, 9]; in the future, a large goal is, among others, to gain insight into the quantum-to-classical transition [10, 7].

Our model is based on References [11, 12]. In this thesis, we investigate a trapped nanoparticle that is coupled to an optical cavity. This cavity is driven by the scattering of photons off the particle. A sketch of the setup can be seen in Figure 2.1. We then find the equilibrium positions by analytic and numerical methods. For the numeric part, we are using the Julia Programming language [13], which provides high performance and high adaptability. Contrary to the research we base this thesis on, we treat this system classically. Establishing a rigorous baseline is a necessary step in identifying quantum mechanical effects in the future. Since we do not explicitly study related quantum phenomena, we may restrict ourselves to the classical description.

In Chapter 2 we will derive the model for a single particle and give the equilibrium positions as a plot of the detuning vs the z values. Using the numeric results we use the Hessian to compute the stability of the system.

In Chapter 3 we extend this model to include N-particles. Here we also introduce necessary conditions and generalize the model.

In Chapter 4 we use this model to find the equilibrium positions for 2 particles. We generate and analyze those results for different phases in the tweezers.

In Chapter 5 we summarize our results and give a brief overview.

Finally, in Chapter 6 we give an outlook on future research options and topics worth analyzing further.

2. Single Particle

2.1. System and Hamiltonian

We begin by considering a dielectric particle with mass m, radius R, and polarizability α. This nanoparticle is trapped by optical tweezers inside an optical cavity of length L. We choose the x-axis to align with the polarization of the laser, the $+z$-axis to be in the propagation direction of the tweezers and the y-axis to align with the cavity axis. The origin is chosen to be at the intensity maximum of the tweezers. In this thesis we assume that $R \ll \lambda$ and we can treat the particle as a point mass compared to the wavelength.

The tweezers have a frequency ω_0, a power of P_t, and a waist W_t at the focus. A sketch of the setup can be seen in Figure 2.1.

With this, our Hamiltonian for a particle at the position $\boldsymbol{r}$ at the time t is

$$H = \frac{||\boldsymbol{p}||^2}{2m} - \frac{1}{2}\boldsymbol{d} \cdot \boldsymbol{E}_{tot}(t, \boldsymbol{r}) + H_c = \frac{\boldsymbol{p}^2}{2m} - \frac{\alpha}{2}||\boldsymbol{E}_{tot}(t, \boldsymbol{r})||^2 + H_c. \tag{2.1}$$

This is because the dipole moment $\boldsymbol{d}$ can be written as

$$\boldsymbol{d} = \alpha \boldsymbol{E}_{tot}. \tag{2.2}$$

Here $\boldsymbol{p}$ is the momentum of the particle, $\boldsymbol{E}_{tot}$ is the total electric field and H_c is the cavity Hamiltonian. This also induces the non-linear self-interaction of the particle with the field.

2.2. Cavity Hamiltonian

In Equation (2.1) H_c represents the energy stored in the cavity and can be written as

$$H_c = \frac{1}{2} \iiint_{\mathbb{R}^3} d^3r \left[\frac{||\boldsymbol{D}||^2}{\epsilon_0} + \frac{||\boldsymbol{\nabla} \times \boldsymbol{A}||^2}{\mu_0} \right], \tag{2.3}$$

with $\boldsymbol{D}$ being the electric displacement field, $\boldsymbol{A}$ being the vector potential, ϵ_0 being the vacuum permittivity and μ_0 the vacuum permeability.

Following Reference [11] the fields in H_c can be decomposed as

$$\boldsymbol{A} = \sum_\beta \boldsymbol{A}_\beta(\boldsymbol{r}) A_\beta(t) \tag{2.4}$$

and

$$\boldsymbol{D} = \sum_\beta \boldsymbol{A}_\beta(\boldsymbol{r}) \epsilon(\boldsymbol{r}) D_\beta(t), \tag{2.5}$$

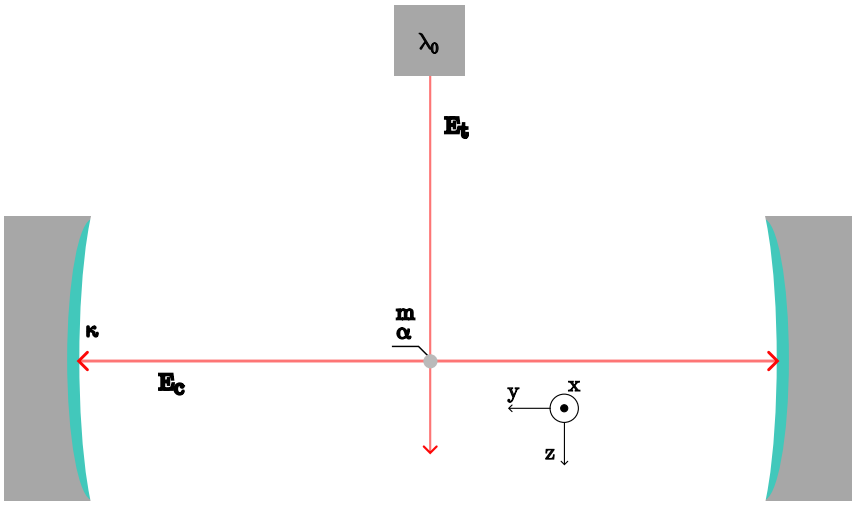


Figure 2.1.: Sketch of the setup

From above we have a Laser with the wavelength λ_0. On the sides we have two optical mirrors that form a Cavity. The gray point describes the dielectric particle,

here β labels the specific mode of the EM field in the presence of the cavity, $\boldsymbol{A}_\beta(\boldsymbol{r})$ being the stationary waves of the vector field and $A_\beta(t) = \exp(i\omega_\beta t)$ being the corresponding time evolution. The displacement field $\boldsymbol{D}_\beta$ is equal to $\epsilon(\boldsymbol{r})\boldsymbol{A}_\beta$ following maxwells equations, with $\epsilon(\boldsymbol{r})$ permittivity of the medium at the position $\boldsymbol{r}$. The steady state solutions remain, while the time evolution may be different, so the time evolution is given by $D_\beta(t)$.

The spacial field $\boldsymbol{A}_\beta(\boldsymbol{r})$ also satisfies

$$(\boldsymbol{A}_\beta, \boldsymbol{A}_\gamma)_{\epsilon(\boldsymbol{r})} = \iiint_{\mathbb{R}^3} d^3r\ \epsilon(\boldsymbol{r})\boldsymbol{A}_\beta(\boldsymbol{r}) \cdot \boldsymbol{A}_\gamma(\boldsymbol{r}) = \delta_{\beta\gamma} \tag{2.6}$$

and the time dependent field satisfies

$$\partial_t A_\beta(t) = i\omega_\beta A_\beta. \tag{2.7}$$

The mode functions are determined such that each partial $\boldsymbol{A}_\beta(\boldsymbol{r})A_\beta(t)$ is a solution of the Helmholtz-Equation

$$\boldsymbol{\nabla} \times \boldsymbol{\nabla} \times \boldsymbol{A}_\beta(\boldsymbol{r})A_\beta(t) = -\frac{1}{c^2}\partial_t^2 \boldsymbol{A}_\beta(\boldsymbol{r})A_\beta(t) \tag{2.8}$$

$$= \frac{\omega_\beta^2}{c^2}\boldsymbol{A}_\beta(\boldsymbol{r})A_\beta(t). \tag{2.9}$$

To simplify the Hamiltonian, we use the algebraic property

$$(\boldsymbol{\nabla} \times \boldsymbol{f}) \cdot (\boldsymbol{\nabla} \times \boldsymbol{g}) = \boldsymbol{f} \cdot (\boldsymbol{\nabla} \times \boldsymbol{\nabla} \times \boldsymbol{g}) - \boldsymbol{\nabla} \cdot (\boldsymbol{f} \times (\boldsymbol{\nabla} \times \boldsymbol{g})), \tag{2.10}$$

which gives the form

$$H_c = \frac{1}{2}\sum_\beta \left[\frac{D_\beta^2(t)}{\epsilon_0} + \epsilon_0\omega_\beta A_\beta^2(t)\right]. \tag{2.11}$$

As shown by Reference [12], the cavity is almost in resonance with the field, so we can approximate the system by limiting it to a single mode, this allows us to write the Hamiltonian as

$$H_c = \frac{D_c^2(t)}{2\epsilon_0} + \frac{1}{2}\epsilon_0\omega_c A_c^2(t). \tag{2.12}$$

We transform to the new coordinates (a, a^*), to simplify our system. The new coordinates are defined by

$$A_c(t) = \sqrt{\frac{\hbar}{2\epsilon_0\omega_c}}\Re\left\{a_c(t)\right\} = \sqrt{\frac{\hbar}{2\epsilon_0\omega_c}}\left(a_c(t) + a_c^*(t)\right) \tag{2.13}$$

and

$$D_c(t) = \sqrt{\frac{\hbar\epsilon_0\omega_c}{2}}\Im\left\{a_c(t)\right\} = -\mathrm{i}\sqrt{\frac{\hbar\epsilon_0\omega_c}{2}}(a_c(t) - a_c^*(t)). \tag{2.14}$$

Substitution gives the cavity Hamiltonian

$$H_c = \hbar\omega_c a_c^*(t)a_c(t). \tag{2.15}$$

The new coordinates do not satisfy the canonical relations. However, they only differ by a constant factor of $\mathrm{i}\hbar$, we may rescale the Poisson brackets to accommodate this. Thus, the new equations of motion are

$$\dot{a}(t) = \{a_c^*, H\} = -\frac{\mathrm{i}}{\hbar}\frac{\partial H}{\partial a_c^*} \tag{2.16}$$

and

$$\dot{a}^*(t) = \{a_c, H\} = +\frac{\mathrm{i}}{\hbar}\frac{\partial H}{\partial a_c}. \tag{2.17}$$

From the equations of motion, we can observe that this is a non-canonical transform, and does change the symplectic structure. The equations of motion can be used to write a new symplectic structure, which is defined by

$$\{\eta_i, \eta_j\} = J_{ij} \tag{2.18}$$

which gives us

$$\underline{\underline{J}} = \frac{1}{\mathrm{i}\hbar}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \tag{2.19}$$

for the subspace $[a_c, a_c^*]$. This allows the equations of motion to be written as

$$\begin{pmatrix} a_c \\ a_c^* \end{pmatrix} = \underline{\underline{J}} \cdot \boldsymbol{\nabla}_{[a_c, a_c^*]} H \tag{2.20}$$

for the subspace.

2.3. Model of the fields

The modeling of this system was done in Reference [12] in a QFT framework, in this thesis, this is adapted into a classical view.

The total electric field can be written as

$$\boldsymbol{E}_{tot}(\boldsymbol{r}) = \boldsymbol{E}_t(\boldsymbol{r}, t) + \boldsymbol{E}_c(\boldsymbol{r}). \tag{2.21}$$

2.3.1. Tweezers field

For optical tweezers, we define them to have a Gaussian structure [14], and thus we can write

$$\boldsymbol{E}_{tot}(\boldsymbol{r},t) = \boldsymbol{e}_x E_t(\boldsymbol{r}) \exp(-\mathrm{i}\omega_0 t) + c.c. \tag{2.22}$$

with the additional scalar function

$$E_t(\boldsymbol{r}) = \frac{E_0}{2} \exp(\mathrm{i}(k_0 z + \phi_t(\boldsymbol{r}))) \frac{W_t}{W(z)} \exp\left(-\left[\frac{y}{A_y W(z)}\right]^2\right) \exp\left(-\left[\frac{x}{A_x W(z)}\right]^2\right) \tag{2.23}$$

where W_t is the waist of the tweezers, $W(z) = W_t\sqrt{1+(z/z_R)^2}$, and the Gouy phase is defined as

$$\phi_t(\boldsymbol{r}) = -\arctan(z/z_R) + \frac{k_0 z}{2}\frac{x^2+y^2}{z^2+z_r^2} \tag{2.24}$$

with $z_R = \pi W_t^2/\lambda_0$ being the Rayleigh range. In general, the beam may be asymmetric in the x and y directions, to include this in the model the parameters of $A_{x,y}$ are introduced.

The field amplitude is written in terms of the power as

$$E_0 = \sqrt{\frac{4P_t}{\pi\epsilon_0 c W_t^2 A_x A_y}}. \tag{2.25}$$

2.3.2. Cavity field

The structure of the Cavity field is already defined by the decomposition we used. From the definition of the displacement field

$$\boldsymbol{E}_c(\boldsymbol{r},t) = \frac{\boldsymbol{D}_c(\boldsymbol{r},t)}{\epsilon_0\epsilon(\boldsymbol{r})} = \frac{\boldsymbol{A}_c(\boldsymbol{r})D_c(t)}{\epsilon_0} \tag{2.26}$$

and our definition of $D_c(t) \propto \Re a_c$, there exists a function $E_c(\boldsymbol{r})$ that allows us to express the total cavity field as

$$\boldsymbol{E}_c(\boldsymbol{r},t) = \boldsymbol{e}_x E_c(\boldsymbol{r}) a_c + c.c.. \tag{2.27}$$

Given the solutions of $\boldsymbol{A}_\alpha(\boldsymbol{r})$ this gives us

$$E_c(\boldsymbol{r}) = E_{0,c}\exp(-(x^2+z^2)/W_c^2)\zeta\cos(k_c y) \in \mathbb{R}, \tag{2.28}$$

and the zero-point field strength of

$$E_{0,c} = \sqrt{\frac{\hbar\omega_c}{2\epsilon_0 V_c}}. \tag{2.29}$$

Similar to the tweezers W_c represents the waist of the Gaussian. The mode Volume is given by $V_c = \pi W_c^2 L_c/4$ with the cavity length L_c, and we also denote $k_c = \omega/c$ as the cavity wave number. For a solution E_t its negative is also a solution to model this we include the parameter of $\zeta = \pm 1$.

2.3.3. Time Averaging

The full expression for $||\boldsymbol{E}_{tot}||$ is

$$||\boldsymbol{E}_{tot}|| = ||\boldsymbol{\varepsilon}_t||^2 + ||\boldsymbol{E}_c||^2 + 2|\boldsymbol{\varepsilon}_t \cdot \boldsymbol{E}_c| \tag{2.30}$$

with

$$||\boldsymbol{\varepsilon}_t||^2 = E_t^2 \exp(-2\mathrm{i}\omega_0 t) + E_t^{*2} \exp(2\mathrm{i}\omega_0 t) + 2|E_t|^2, \tag{2.31}$$

$$||\boldsymbol{E}_c||^2 = E_c^2 a_c^2 + E_c^2 a_c^{*2} + 2E_c^2|a_c|^2 = E_c^2 \alpha_c \alpha_c^* + E_c^2 \left(\alpha_c^2 \exp(-2\mathrm{i}\omega_0 t) + c.c.\right), \tag{2.32}$$

and

$$|\boldsymbol{\varepsilon}_t \cdot \boldsymbol{E}_c| = E_c(E_t \exp(-\mathrm{i}\omega_0 t) + c.c.)(\alpha_c \exp(-\mathrm{i}\omega_0 t) + c.c.). \tag{2.33}$$

Here we define

$$a_c := \alpha_c \exp(-\mathrm{i}\omega_0 t) \tag{2.34}$$

To eliminate very fast-changing terms we take the time average of the time $T = \pi/\omega_0$. The fast changing terms vanish as we integrate over a period and for the slower changing terms the time averaging acts like a derivative and cancels the integral. This yields

$$||\boldsymbol{E}_{tot}(\boldsymbol{r})||^2 = 2\left(|E_t(\boldsymbol{r})|^2 + E_c^2(\boldsymbol{r}) a_c^* a_c + [a_c E_c E_t^* \exp(i\omega_0 t) + c.c.]\right). \tag{2.35}$$

2.4. Equations of Motion

Using the above modeling and approximations the Hamiltonian can be written as

$$H = \frac{||\boldsymbol{p}||^2}{2m} + \hbar\omega_c a_c^* a_c - \alpha\left[|E_t(\boldsymbol{r})|^2 + E_c^2(\boldsymbol{r}) a_c^* a_c + [a_c E_c E_t^* \exp(\mathrm{i}\omega_0 t) + c.c.]\right]. \tag{2.36}$$

Applying Equation (2.16) leads to

$$\dot{a}_c(t) = -\mathrm{i}\left(\omega_c - \frac{\alpha}{\hbar} E_c^2(\boldsymbol{r})\right) a_c + \frac{\mathrm{i}\alpha}{\hbar} E_t(\boldsymbol{r}) E_c(\boldsymbol{r}) \exp(-\mathrm{i}\omega_0 t). \tag{2.37}$$

We use the canonical transform

$$a_c(t) = \alpha_c(t) \exp(-\mathrm{i}\omega_0 t) \tag{2.38}$$

to transform the Hamiltonian to be time independent. Applying the transform gives

$$\dot{\alpha}_c(t) = -\mathrm{i}\underbrace{\left(\omega_c - \omega_0 - \frac{\alpha}{\hbar} E_c^2(\boldsymbol{r})\right)}_{:=\delta_c} \alpha_c(t) + \frac{\mathrm{i}\alpha}{\hbar} E_t(\boldsymbol{r}) E_c(\boldsymbol{r}), \tag{2.39}$$

and with this transform the Hamiltonian can be expressed as

$$H = \frac{||\boldsymbol{p}||^2}{2m} + \hbar\Delta\alpha_c^*\alpha_c - \alpha\left[|E_t(\boldsymbol{r})|^2 + E_c^2(\boldsymbol{r})\alpha_c^*\alpha_c + [\alpha_c E_c E_t^* + c.c.]\right]. \tag{2.40}$$

This expression uses the detuning Δ defined as

$$\Delta := \omega_c - \omega_0. \tag{2.41}$$

Our goal is to determine the equilibrium values for all the classical amplitudes, $\boldsymbol{r}$ and α_c. At equilibrium the time derivative of these generalized coordinates needs to be set to 0. Applying this to the cavity field yields

$$\alpha_c^{eq} = \frac{\alpha}{\hbar} \frac{E_t(\boldsymbol{r}^{eq}) E_c(\boldsymbol{r}^{eq})}{\delta_c - i\frac{\kappa}{2}}. \tag{2.42}$$

The extra term of $-i\kappa/2$ is included to model the loss of light, due to imperfect mirrors, Reference [12] shows that this gives a good description of damping. The additional term is also needed to avoid divergence at $\Delta = 0$.

Applying the equilibrium condition to the particle position gives

$$\ddot{\boldsymbol{r}} \propto \dot{\boldsymbol{p}} = -\boldsymbol{\nabla}_{\boldsymbol{r}} H = +\alpha \boldsymbol{\nabla}_{\boldsymbol{r}} |E_t(\boldsymbol{r}) + \alpha_c(t) E_c(\boldsymbol{r})|^2 \stackrel{!}{=} \boldsymbol{0} \tag{2.43}$$

or equivalently

$$\boldsymbol{\nabla} |E_t(\boldsymbol{r}) + \alpha_c^{eq}(\boldsymbol{r}^{eq}) E_c(\boldsymbol{r})|^2 |_{\boldsymbol{r}=\boldsymbol{r}^{eq}} \stackrel{!}{=} \boldsymbol{0}. \tag{2.44}$$

2.5. Finding the Extremes

The identity $\boldsymbol{\nabla}|h|^2 = 2\Re\{h^* \boldsymbol{\nabla} h\}$ can be utilized to rewrite the expression as

$$\Re\{(E_t^*(\boldsymbol{r}) + \alpha_c^{eq*} E_c(\boldsymbol{r}))(\boldsymbol{\nabla} E_t(\boldsymbol{r}) + \alpha_c^{eq} \boldsymbol{\nabla} E_c(\boldsymbol{r}))\} \stackrel{!}{=} 0. \tag{2.45}$$

We can simplify the expression further, by using the logarithmic derivatives

$$\boldsymbol{f}(\boldsymbol{r}) = \boldsymbol{\nabla} \ln E_t(\boldsymbol{r}) \tag{2.46}$$

$$\text{and } \boldsymbol{g}(\boldsymbol{r}) = \boldsymbol{\nabla} \ln E_c(\boldsymbol{r}). \tag{2.47}$$

With those definitions, the Equation (2.45) can be rewritten as

$$\Re\{(E_t^*(\boldsymbol{r}^{eq}) + \alpha_c^{eq*} E_c(\boldsymbol{r}^{eq}))\,(\boldsymbol{f}(\boldsymbol{r}^{eq}) E_t(\boldsymbol{r}^{eq}) + \alpha_c^{eq} \boldsymbol{g}(\boldsymbol{r}) E_c(\boldsymbol{r}))\} \tag{2.48}$$

$$= |E_t|^2 \Re\left\{\left(1 + \frac{\alpha}{\hbar} \frac{E_c^2}{\delta_c + i\frac{\kappa}{2}}\right)\left(\boldsymbol{f} + \frac{\alpha}{\hbar} \frac{E_c^2}{\delta_c - i\frac{\kappa}{2}} \boldsymbol{g}\right)\right\}(\boldsymbol{r}^{eq}) \tag{2.49}$$

$$\stackrel{!}{=} 0. \tag{2.50}$$

From the model of the tweezers in Section 2.3 it is obvious that $|E_t| > 0$ for all positions $||\boldsymbol{r}|| < \infty$, as such, it can be removed without missing any roots.

Thus, the expression can be written as

$$\Re\left\{\left(1 + \gamma\left(\delta_c - \mathrm{i}\frac{\kappa}{2}\right)\right)\left(\boldsymbol{f} + \gamma\left(\delta_c + \mathrm{i}\frac{\kappa}{2}\right)\boldsymbol{g}\right)\right\} \stackrel{!}{=} 0. \tag{2.51}$$

with

$$\gamma := \frac{\alpha}{\hbar} \frac{E_c^2(\boldsymbol{r}^{eq})}{\delta_c^2 + \frac{\kappa^2}{4}} \in \mathbb{R}. \tag{2.52}$$

Using the property that $\Re\{z_1 z_2\} = \Re z_1 \Re z_2 - \Im z_1 \Im z_2$ this becomes

$$(1 + \gamma\delta_c)\,(\Re \boldsymbol{f} + \boldsymbol{g}\gamma\delta_c) + \gamma\frac{\kappa}{2}\left(\Im \boldsymbol{f} + \boldsymbol{g}\gamma\frac{\kappa}{2}\right) \stackrel{!}{=} 0. \tag{2.53}$$

This expression cannot be simplified further in vector form. Thus, we continue to find solutions for the individual components.

2.5.1. x-Component

Based on the fields in equations (2.23) and (2.28) the x-component of the logarithmic gradients are

$$f_x = x\left[\mathrm{i}\frac{k_0 z}{z^2+z_R^2} - \frac{2}{A_x^2 W^2(z)}\right] \tag{2.54}$$

and

$$g_x = -\frac{2x}{W_c^2}. \tag{2.55}$$

Substitution into Equation (2.53) gives

$$x\left[-2(1+\gamma\delta_c)\left(\frac{1}{2A_x^2W^2(z)} + \frac{1}{W_c^2}\gamma\delta_c\right) + \gamma\frac{\kappa}{2}\left(\frac{k_0 z}{z^2+z_R^2} - \gamma\frac{\kappa}{2}\frac{2}{W_c^2}\right)\right] \stackrel{!}{=} 0. \tag{2.56}$$

This equation has a trivial solution at $x = 0$. It follows from the symmetry of the Problem that no other value of x will be a solution.

We also define $h_x := \frac{1}{x}\partial_x H$, from the above results, it is also obvious that h_x does not diverge at $x = 0$.

2.5.2. y-Component

Analogous to the logarithmic gradients of the x component, based on the definition of fields in Equations (2.23) and (2.28), the logarithmic gradients of the y-component are

$$f_y = y\left[\mathrm{i}\frac{k_0 z}{z^2+z_R^2} - \frac{2}{A_y^2 W^2(z)}\right] \tag{2.57}$$

and

$$g_y = -k_c\tan(k_c y) \tag{2.58}$$

Substitution into Equation (2.53) gives

$$\begin{aligned} y\Bigg[-2(1+\gamma\delta_c)&\left(\frac{1}{2A_y^2W^2(z)} + \frac{k_c\tan(k_c y)}{y}\frac{1}{W_c^2}\gamma\delta_c\right) \\ &+ \gamma\frac{\kappa}{2}\left(\frac{k_0 z}{z^2+z_R^2} - \frac{k_c\tan(k_c y)}{y}\gamma\frac{\kappa}{2}\frac{2}{W_c^2}\right)\Bigg] \stackrel{!}{=} 0. \end{aligned} \tag{2.59}$$

Which results in a trivial solution of $y = 0$. Similar to the x component, symmetry forbids the existence of other solutions.

Also note that since $\tan(x) = x + \mathcal{O}(x^3)$, the term $\tan(k_c y)/y|_{y=0}$ is a removable singularity. Similarly to the x-component we define $h_y(\boldsymbol{r}) := \frac{1}{y}\partial_y H$ and similarly h_y does not diverge at $y = 0$.

2.5.3. z-Component

Since the solutions for x and y are known, we can only consider the case where they are satisfied. The logarithmic gradients for the z component are

$$f_z = \left(\mathrm{i} \left(k_0 - \frac{z_R}{z^2 + z_R^2} \right) - \frac{z}{z^2 + z_R^2} \right) \tag{2.60}$$

and

$$g_z = -\frac{2z}{W_c^2}. \tag{2.61}$$

With those components and Equation (2.53), we get the expression

$$L_z := -z(1 + \gamma\delta_c) \left(\frac{1}{z^2 + z_R^2} + \frac{2}{W_c^2}\gamma\delta_c \right) - \gamma\frac{\kappa}{2} \left(-k_0 + \frac{z_R}{z^2 + z_R^2} + 2\frac{z}{W_c^2}\gamma\frac{\kappa}{2} \right) \stackrel{!}{=} 0. \tag{2.62}$$

With this, the solution manifold is 1 dimensional and fully characterized by the z coordinate.

2.6. Analytical Special cases

Before we consider the numeric solutions, we take a look at some special cases for the damping κ. This gives us an understanding of our system in these limits and helps us check if the numeric computations follow this behavior.

2.6.1. Case $\kappa \to \infty$

Since γ is proportional to κ^{-2}, it immediately follows that $\gamma \to 0$ and $\gamma\kappa \to 0$. Which simplifies our expression to

$$\frac{z}{z^2 + z_R^2} = 0, \tag{2.63}$$

which gives us the trivial solution of $z = 0$.

This is also expected, because if the damping is very high the cavity field can be neglected and only the tweezers remain.

2.6.2. Case $\kappa \to 0$

For this case we require that $\delta_c \neq 0$, to avoid divergence.

Our α_c^{eq} simplifies to

$$\alpha_c^{eq} = \frac{\alpha}{\hbar}\frac{E_t E_c}{\delta_c} \tag{2.64}$$

and similarly γ simplifies to

$$\gamma = \frac{\alpha}{\hbar}\frac{E_c^2}{\delta_c^2}. \tag{2.65}$$

And we are left with only the first term of our expression

$$z(1 + \gamma\delta_c) \left(\frac{1}{z^2 + z_R^2} + \frac{2}{W_c^2}\gamma\delta_c \right) \stackrel{!}{=} 0, \tag{2.66}$$

which also gives us the trivial solution of $z = 0$.

2.6.3. Case $\Delta \to \infty$

This case is similar to the $\kappa \to \infty$ case 2.6.1. Here $\gamma \propto \Delta^{-2}$, which also implies that $\gamma \to 0$. Since $\delta_c \sim \Delta$, it follows that $\delta_c \gamma \to 0$ as well. These properties give us the same equilibrium position

$$\frac{z}{z^2 + z_R^2} = 0 \tag{2.67}$$

which also has a trivial solution at $z = 0$.

2.7. Numerical Computation

Now we take a look at the numerics and the evaluation of the equilibrium solutions. The numerical calculation is done using the Julia programming language (Version `1.12`) [13]. The basic setup consists of implementing our model in Julia and writing the expression in Equation (2.62) as a function $L(z; \Delta, \kappa)$.

To find the roots of L, we create two intervals for z and for Δ. For each z, we iterate over our Δ values and check if a root is between one detuning value and the next, i.e., a sign change in the function occurs. Since L is continuous, the intermediate value theorem guarantees the existence of a root in this interval. If there is a root in such an sub interval, we use `Roots.jl` (Version `2.2`) [15] to refine the interval and get an exact result. Since `Roots.jl` gives roots up to the numerical precision of the respective type, the errors will be in the order of $\sim eps(\texttt{Float64}) \simeq 2.22e-16$ which are for our purposes negligible.

In the following calculation, we choose the interval for z to be logarithmic and the interval for Δ to be linear. The z values are in the interval from $10^{-2}\mu$m to 50μm and the Δ values are in the interval from 0kHz to 800 2πkHz.

We can also vary our κ parameter to examine our system's behavior under different damping rates. Such a plot can be seen in Figure 2.2. The parameters for this figure can be seen in Table 2.1. From the plots, it can be observed that the curve for the detuning $\Delta(z)$ with fixed κ follows the characteristic inverse "S" shape that can be observed in cubic polynomials. Also, notably this gives us configurations with multiple equilibrium positions. In the plot, it can be seen that, with an increasing κ, the curve gets flatter and the vertices get further apart as compared to less κ. This implies a decreased sensitivity to detuning, which manifests as a wider spacing between the vertices.

Extrapolating this for $\kappa \to \infty$, the curve would collapse into a single line at $z = 0$. This behavior is consistent with our analytical prediction in Section 2.6. Conversely, for $\kappa \to 0$ we would get that for every value of Δ there would only be a single value for z. If we take this even further, the curve would tend to infinity at $\Delta = 0$ and go to $z = 0$. This also matches our prediction of this case in Section 2.6.

We can also observe that the solutions tend to 0 for larger detunings, which also matches the prediction of Section 2.6.

The limiting behaviors confirm the validity of our calculation. These results have never been predicted in literature. Since one typically either assumes larger values for κ or has only access to larger values of κ in experiments.

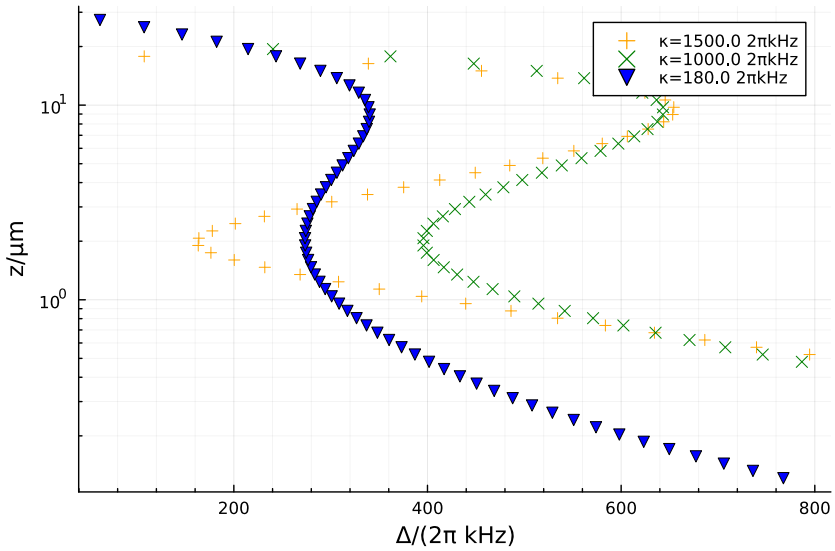


Figure 2.2.: Plot of the equilibrium positions as a function of z and Δ for multiple values of κ, with the parameters of Table 2.1

Laser Parameter	Value	
λ_0	1150	nm
P_t	0.5	W
W_t	1	μm
$A_x = A_y$	1	

Particle Parameter	Value	
ρ	2200	kg m^{-3}
ϵ	2.07	
R	100	nm

Cavity Parameter	Value	
W_c	20	μm
L_c	19.8	mm
ζ	1	

Table 2.1.: Parameters of the calculation for one particle.

The parameters are taken from Reference [12]

2.8. Stability

For any experimental use case of this system, it would be important to know the stability of the equilibrium positions. In this section, we will derive the stability of the generated equilibrium conditions.

We define $\boldsymbol{\eta} := [x, p_x, y, p_y, z, p_z, \alpha_c, \alpha_c^*]$, with this Hamilton's equation are

$$\dot{\boldsymbol{\eta}} = \underline{\underline{J}} \cdot \boldsymbol{\nabla}_{\boldsymbol{\eta}} H - \underline{\underline{K}} \cdot \boldsymbol{\eta}. \tag{2.68}$$

with $\underline{\underline{K}}$ modeling the damping

$$\underline{\underline{K}} = \underline{\underline{Diag}}\left(0, \ldots, 0, -\frac{\kappa}{2}, -\frac{\kappa}{2}\right). \tag{2.69}$$

To analyze the stability we can linearize the equation of motion around the equilibrium position $\boldsymbol{\eta}_0$

$$\begin{aligned}
\dot{\boldsymbol{\eta}} &= \underline{\underline{J}} \cdot \boldsymbol{\nabla}_\eta H - \underline{\underline{K}} \cdot \boldsymbol{\eta} && (2.70)\\
&= \underline{\underline{J}} \cdot (\boldsymbol{\nabla}_\eta \boldsymbol{\nabla}_\eta H(\boldsymbol{\eta}_0) \cdot \boldsymbol{\eta}) + \mathcal{O}(||\boldsymbol{\eta}||^2) - \underline{\underline{K}} \cdot \boldsymbol{\eta} && (2.71)\\
&\approx \underline{\underline{J}} \cdot \underline{\underline{\text{Hess}}}_\eta H \cdot \boldsymbol{\eta} - \underline{\underline{K}} \cdot \boldsymbol{\eta} && (2.72)\\
&:= \underline{\underline{S}} \cdot \boldsymbol{\eta}, && (2.73)
\end{aligned}$$

Note that this $\underline{\underline{J}}$ is not the canonical symplectic form usually found in classical mechanics, which is a result of our non-canonical transform in Section 2.2.

To analyze the stability, we need to check the Eigenvalues of the Matrix $\underline{\underline{S}}$. The results will be superpositions of $\boldsymbol{v}_i \cdot \exp(\lambda_i \cdot t)$ and for the system to be stable none of those may diverge to infinity, i.e. the Eigenvalues need to satisfy $\Re\lambda_i \leq 0$.

Every first-order partial derivative of H with x or y can be expressed as $\partial_i H = r_i h_i(\boldsymbol{r}),\ i \in \{x, y\}$. Since every equilibrium has $x = 0 = y$, we can argue that every partial derivative of $\partial_j \partial_i H = r_i \partial_j h_i(\boldsymbol{r}) = 0,\ i \in \{x, y\},\ j \in \{x, y, z, \alpha_c, \alpha_c^*\},\ j \neq i$ at $x = 0 = y$. Furthermore, since the momentum only appears in the kinetic term, the every momentum derivative is $\partial_{p_i} H = p_i/m$. In this ordering, the Hessian of the Hamiltonian can be written using a diagonal block matrix

$$\underline{\underline{\text{Hess}}} H = \begin{pmatrix} \underline{\underline{X}} & & \\ & \underline{\underline{Y}} & \\ & & \underline{\underline{C}} \end{pmatrix}. \tag{2.74}$$

The matrices $\underline{\underline{X}} \in \mathbb{R}^{2\times 2}$ and $\underline{\underline{Y}} \in \mathbb{R}^{2\times 2}$ are the Hessians of the x, p_x and y, p_y subspaces respectively, the matrix $\underline{\underline{C}} \in \mathbb{C}^{4\times 4}$ is the Hessian with respect to the remaining coordinates.

Our symplectic structure is

$$\underline{\underline{J}} = \begin{pmatrix} \underline{\underline{j}} & & & \\ & \underline{\underline{j}} & & \\ & & \underline{\underline{j}} & \\ & & & \frac{1}{\mathrm{i}\hbar}\underline{\underline{j}} \end{pmatrix},\ \underline{\underline{j}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{2.75}$$

This gives the expression

$$\underline{\underline{J}} \cdot \underline{\underline{\text{Hess}}} H = \begin{pmatrix} 0 & \frac{1}{m} & & & \\ -\partial_x^2 H & 0 & & & \\ & & 0 & \frac{1}{m} & \\ & & -\partial_y^2 H & 0 & \\ & & & & \underline{\underline{B}} \end{pmatrix}, \tag{2.76}$$

with

$$\underline{\underline{B}} = \begin{pmatrix} 0 & \frac{1}{m} & 0 & 0 \\ -\partial_z^2 H & 0 & -\partial_z \partial_{\alpha_c} H & -\partial_z \partial_{\alpha_c^*} H \\ \frac{1}{\mathrm{i}\hbar}\partial_z \partial_{\alpha_c^*} H & 0 & \frac{1}{\mathrm{i}\hbar}\partial_{\alpha_c} \partial_{\alpha_c^*} H & 0 \\ -\frac{1}{\mathrm{i}\hbar}\partial_z \partial_{\alpha_c} H & 0 & 0 & -\frac{1}{\mathrm{i}\hbar} H \partial_{\alpha_c} \partial_{\alpha_c^*} H \end{pmatrix}. \tag{2.77}$$

By Laplace expansion of the by λ shifted matrix, it is observed that the x and y components decouple completely. This means that we can get the eigenvalues of each block of the matrix individually. Formally this is stated as

$$\left|\underline{\underline{j}} \cdot \underline{\underline{X}} - \mathbb{1}\lambda\right| \cdot \left|\underline{\underline{j}} \cdot \underline{\underline{Y}} - \mathbb{1}\lambda\right| \cdot \left|\underline{\underline{B}} - \mathbb{1}\lambda\right| \stackrel{!}{=} 0. \tag{2.78}$$

We get that Eigenvalues of the decoupled x and y components are

$$\lambda = \pm \mathrm{i}\sqrt{\frac{\partial_i^2 H}{m}}\ , i \in \{x, y\}. \tag{2.79}$$

It is trivial to see that the stability criterion for those components is $\partial_i^2 H > 0$. From our computation of the gradients in Sections 2.5.1 and 2.5.2 the derivatives take the form $\partial_i H = r_i h_i(\boldsymbol{r})$, where we call $r_x = x$ and $r_y = y$. The second derivative can be computed to $\partial_i^2 H = h_i(\boldsymbol{r})$, which is given by the analytic computations. We check the stability of $\underline{\underline{X}}$ and $\underline{\underline{Y}}$ from the analytic results, and we compute the Hessian with respect to $[z, p_z, \alpha_c, \alpha_c^*]$ numerically. Since we cannot compute the numeric derivatives of a complex variable numerically, we need to transform the Hessian of our matrix, the formalism for this is described in the Appendix in Section B.1. With this transform, we can take the numeric derivative of our Hamiltonian. We do this using the `ForwardDiff.jl` (Version `1.3`) [16] package. The result of the stability check can be seen in Figure 2.3.

We can observe that the system is always stable, in the considered parameter regime.

An interesting special case for the stability analysis is if α_c reaches its equilibrium position instantaneously. The Analysis for this system is done in the Appendix under Section B.2. However, the Cavity needs several round-trips between the mirrors to reach its equilibrium state and thus, this approximation does not hold in our system.

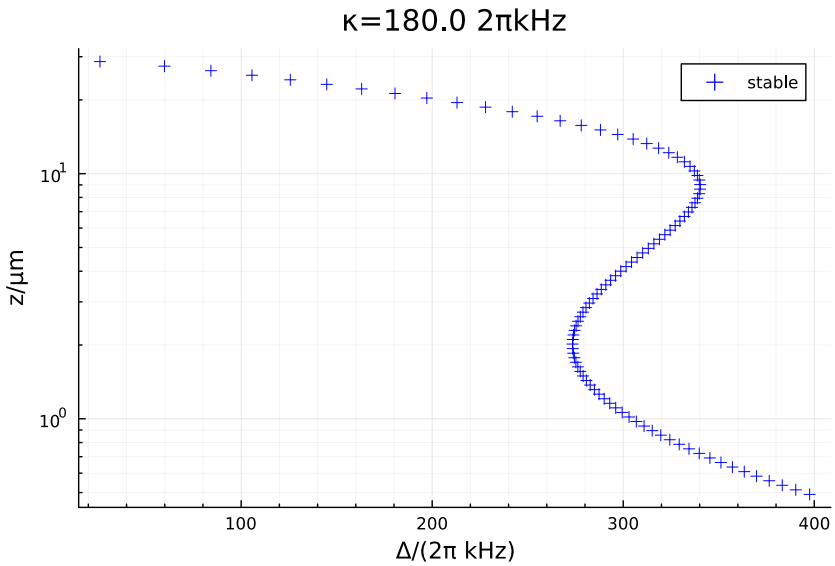


Figure 2.3.: Plot of the equilibrium position at $\kappa = 180.0\ 2\pi kHz$ showing the stability of the equilibrium points with the parameters of Table 2.1

3. N-Particles

In this chapter, we will generalize the system from Chapter 2 to N-particles. For this, we will consider the same setup as in the 1-Particle case, but instead of a single particle and an optical tweezers, we consider N Particles with N optical tweezers arranged along the y-Axis. A sketch of the setup can be seen in Figure 3.1

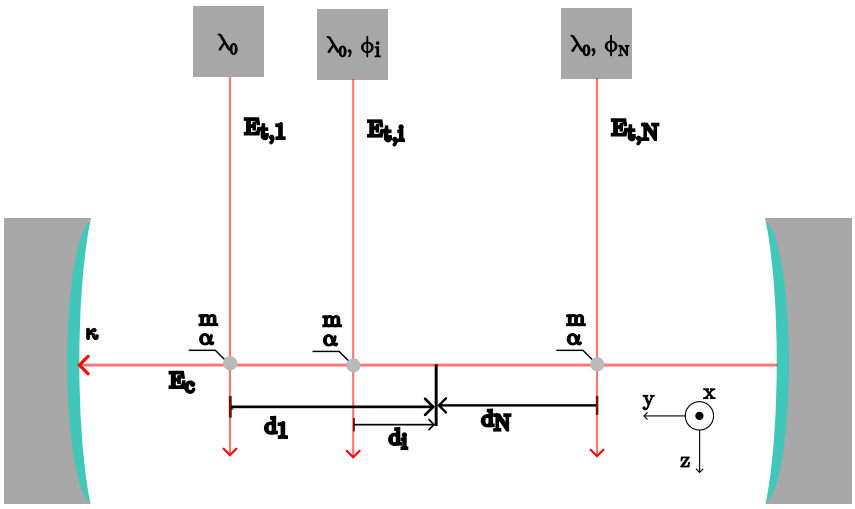


Figure 3.1.: Setup for N-particles

Here we have N-Lasers of wavelength λ_0 and a relative phase ϕ_i. On the sides are the optical mirrors and at the focal point of each laser is a particle.

3.1. Adapting the Modeling

A lot of the formalism can be carried over from the 1-Particle case. To get the correct Hamiltonian, we must sum all kinetic terms and all dipole interactions. There is still only a single Cavity, and from that H_C will remain unchanged.

$$H = \sum_i \left[\frac{||\boldsymbol{p}_i^2||}{2m_i} - \frac{1}{2}\alpha_i ||\boldsymbol{E}_{tot}(\boldsymbol{r}_i)||^2 \right] + \hbar\omega_c a_c^* a_c \tag{3.1}$$

with

$$\boldsymbol{E}_{tot}(\boldsymbol{r}) = \boldsymbol{E}_c(\boldsymbol{r}) + \sum_i \varepsilon_i(\boldsymbol{r}). \tag{3.2}$$

We assume all optical tweezers $\boldsymbol{\varepsilon}_i(\boldsymbol{r})$ to be modeled by the same Gaussian as in the 1-Particle case. Each optical tweezers gains two additional degrees of freedom, the position along the y-axis can be varied, which we will call $\boldsymbol{R}_i = d_i \boldsymbol{e}_y$ and the phase, which we will denote with φ_i. We also assume that the rest of the parameters are assumed to be equivalent. With this, we write our new tweezers as

$$\boldsymbol{\varepsilon}_i(\boldsymbol{r}) = \boldsymbol{e}_x \left[E_{t,i}(\boldsymbol{r}) \exp(-\mathrm{i}\omega_0 t) + c.c.\right] \tag{3.3}$$

$$E_{t,i}(\boldsymbol{r}) = E_t\left(\boldsymbol{r} - \boldsymbol{R}_i\right) \exp(\mathrm{i}\varphi_i). \tag{3.4}$$

Since any global phase is not detectable, we may choose $\varphi_1 = 0$ without loss of generality.

To make the modeling easier, we assume that the distances between the intensity maximums of the tweezers satisfy $|d_i - d_j| \gg W_t \; \forall \; 1 \leq i, j < N, i \neq j$, such that the optical tweezers only act on their respective particle. Due to this approximation, the particles will only couple through the cavity field and have no interaction otherwise. Under this assumption, the total electric field becomes

$$\boldsymbol{E}_{tot}(\boldsymbol{r}_i) \approx E_c(\boldsymbol{r}_i) + \boldsymbol{\varepsilon}_i(\boldsymbol{r}_i). \tag{3.5}$$

Time averaging the system simplifies our Hamiltonian, in complete analogy to the 1-Particle case, we get

$$\begin{aligned} H = &- \sum_i \left[\frac{\alpha}{2}\left[|E_{t,i}(\boldsymbol{r}_i)|^2 + E_c^2(\boldsymbol{r}_i) a_c^* a_c + \left[E_c(\boldsymbol{r}_i) E_{t,i}^*(\boldsymbol{r}_i) a_c \exp(\mathrm{i}\omega_0 t) + c.c.\right]\right]\right] \\ &+ \sum_i \frac{1}{2m} ||\boldsymbol{p}_i||^2 + \hbar\omega_c a_c^* a_c. \end{aligned} \tag{3.6}$$

3.2. Equations of Motion

This Hamiltonian can be used to get the time evolution of a_c

$$\dot{a}_c = -\mathrm{i}\left(\omega_c - \frac{\alpha}{\hbar}\sum_i E_c^2(\boldsymbol{r}_i)\right) a_c + \frac{\mathrm{i}\alpha}{\hbar}\sum_i E_c(\boldsymbol{r}_i) E_{t,i}(\boldsymbol{r}_i) \exp(-\mathrm{i}\omega_0 t). \tag{3.7}$$

The same transformation as with the 1-Particle case can be used to remove the explicit time dependence. Setting the time derivative to zero and simplifying yields

$$\alpha_c^{eq} = \frac{\alpha}{\hbar} \frac{\sum_i E_c(\boldsymbol{r}_i) E_{t,i}(\boldsymbol{r}_i)}{\omega_c - \omega_0 - \frac{\alpha}{\hbar}\sum_i E_c^2(\boldsymbol{r}_i)}. \tag{3.8}$$

Damping needs to be added the same way by adding a term of $-i\kappa/2$ to the denominator. Similarly to the 1-Particle case we also define

$$\delta_c := \omega_c - \omega_0 - \frac{\alpha}{\hbar}\sum_i E_c^2(\boldsymbol{r}_i) = \Delta - \frac{\alpha}{\hbar}\sum_i E_c^2(\boldsymbol{r}_i). \tag{3.9}$$

Which gives almost the same form as the 1-Particle case

$$\alpha_c^{eq} = \frac{\alpha}{\hbar} \frac{\sum_i E_c(\boldsymbol{r}_i) E_{t,i}(\boldsymbol{r}_i)}{\delta_c - i\frac{\kappa}{2}} \tag{3.10}$$

The new condition for stability is

$$\dot{\boldsymbol{p}}_i = -\boldsymbol{\nabla}_i H = +\alpha \boldsymbol{\nabla}_i |\alpha_c E_c(\boldsymbol{r}_i) + E_{t,i}(\boldsymbol{r}_i)|^2 \stackrel{!}{=} \boldsymbol{0}. \tag{3.11}$$

$\boldsymbol{\nabla}_i$ being the spatial derivative of the i-th Particle.

3.3. Finding the Extremes

We will use logarithmic gradients again. The definition for $\boldsymbol{g}$ does not change and can simply be reused from Section 2.5. The definition for $\boldsymbol{f}$ needs to be adapted to be

$$\boldsymbol{f}_i(\boldsymbol{r}_i) := \boldsymbol{\nabla}_i \ln E_{t,i}(\boldsymbol{r}_i) \Leftrightarrow \boldsymbol{\nabla}_i E_{t,i}(\boldsymbol{r}_i) = \boldsymbol{f}_i(\boldsymbol{r}_i) E_{t,i}(\boldsymbol{r}_i) \Leftrightarrow \boldsymbol{f}_i(\boldsymbol{r}_i) = \boldsymbol{f}(\boldsymbol{r}_i - \boldsymbol{R}_i). \tag{3.12}$$

We use the property of $\partial|f|^2 = 2\Re\{f^*\partial f\}$ to get the equilibrium condition

$$2\Re\left\{\left[\alpha_c^* E_c(\boldsymbol{r}_j) + E_{t,j}(\boldsymbol{r}_j)\right]\left[\alpha_c \boldsymbol{g}(\boldsymbol{r}_j) E_c(\boldsymbol{r}_j) + \boldsymbol{f}_j(\boldsymbol{r}_j) E_{t,j}(\boldsymbol{r}_j)\right]\right\} \stackrel{!}{=} 0. \tag{3.13}$$

Since we introduced more complex degrees of freedom, we cannot simplify this expression further. However, this form is already sufficient to analyze its components, which we will do in the following sections.

3.3.1. x-Component

Based on the definitions in Equations (3.3) and (2.28), the logarithmic gradients for the x component are

$$f_{x,j}(\boldsymbol{r}_j) = x\left[\mathrm{i}\frac{k_0 z_j}{z_j^2 + z_R^2} - \frac{2}{A_x^2 W^2(z_j)}\right] \tag{3.14}$$

and

$$g_x(\boldsymbol{r}_j) = -\frac{2x_j}{W_c^2}. \tag{3.15}$$

In the second multiplicative term, we can factor out an x_j

$$x_j\left[-\alpha_c \frac{2}{W_c^2} E_c(\boldsymbol{r}_j) + \left[\mathrm{i}\frac{k_0 z_j}{z_j^2 + z_R^2} - \frac{2}{A_x^2 W^2(z_j)}\right] E_{t,j}(\boldsymbol{r}_j)\right]. \tag{3.16}$$

Since x_j is real, it can be factored out of the $\Re$ operation and the condition is satisfied by $x_j = 0$. Which also can be expected since in the 1 particle case, the solution was $x = 0$ independent of y and z and the shift does not affect the x direction in any way.

3.3.2. y-Component

Based on the definitions in Equations (3.3) and (2.28), the logarithmic gradients are

$$f_{y,j}(\boldsymbol{r}_j) = (y_j - d_j)\left[\mathrm{i}\frac{k_0 z_j}{z_j^2 + z_R^2} - \frac{2}{A_y^2 W^2(z_j)}\right] \tag{3.17}$$

and

$$g_y(\boldsymbol{r}_j) = -k_c \tan(k_c y) \tag{3.18}$$

If we require $y_i = d_i$ and $\tan k_c y_i = 0$, both $f_{y,i}$ and g_y are zero. Substituting this into the second multiplicative term in Equation (3.13) gives

$$[0 \cdot \alpha_c E_c(\boldsymbol{r}_j) + 0 \cdot E_{t,j}(\boldsymbol{r}_j)] = 0, \tag{3.19}$$

which also immediately satisfies the condition.

With this, we can conclude that the particle will stay at $y_i - d_i = 0$ if $\tan k_c d_i = 0$ is satisfied. This condition adds a constraint on where we can place the tweezers. For this thesis, we will only consider cases where this condition is satisfied.

3.3.3. z-Component

To simplify this condition, we use the previous conditions. ($x = 0$ and $y_i = d_i$) Under those, we can write the special cases of our tweezers field as

$$E_{t,i}(\boldsymbol{r}_i) = E_t(z\boldsymbol{e}_z)\exp(\mathrm{i}\phi_i). \tag{3.20}$$

This gives us the logarithmic gradients in z as

$$f_{z,j}(\boldsymbol{r}_j) = f_z(z_j\boldsymbol{e}_z) = \left[\mathrm{i}\left(k_0 - \frac{z_R}{z_j^2 + z_R^2}\right) - \frac{z_j}{z_j^2 + z_R^2}\right] \tag{3.21}$$

and

$$g_z(\boldsymbol{r}_j) = -\frac{2z_j}{W_c^2}. \tag{3.22}$$

We may substitute this finding into Equation (3.13), and we get a condition that depends on the particle positions in z_j. Note that since we assumed the tweezers would not affect each other, the coupling to the other particle positions originates only from the equilibrium condition of the cavity field $\alpha_c^{eq}(z_1, z_2, ..., z_N)$.

For N particles, we get N coupled conditions. We can gather all of those conditions and write them as one multi variable problem. To do this, we define

$$\boldsymbol{L}(z_1, z_2, ..., z_N) = \boldsymbol{L}(\boldsymbol{z}) = 0,\ \boldsymbol{L} : \mathbb{R}^N \to \mathbb{R}^N \tag{3.23}$$

with each component representing the partial derivative of a different particle's position, or written as a formula

$$L_i(\boldsymbol{z}) := \alpha \boldsymbol{\nabla}_i |\alpha_c^{eq} E_c(\boldsymbol{r}_i) + E_{t,i}(\boldsymbol{r}_i)|^2 \tag{3.24}$$

$$= 2\alpha \Re \left\{ [\alpha_c^{eq*} E_c(\boldsymbol{r}_i) + E_{t,i}(\boldsymbol{r}_i)] \left[\alpha_c^{eq} g_z(\boldsymbol{r}_i) E_c(\boldsymbol{r}_i) + f_{z,i}(\boldsymbol{r}_i) E_{t,i}(\boldsymbol{r}_i)\right] \right\}. \tag{3.25}$$

3.4. Numerical Computation

For the numeric computation, we use the Package `NLsolve.jl` (Version `4.5.1`), with a trust region Newton style algorithm as described by Dennis and Schnabel [17]. This method reduces divergence to ∞ for the solutions, which is prone to happen using the standard Newton algorithm.

Similar to the standard Newton algorithm, this method requires starting guesses. For those, we use the N-fold Cartesian product of the solutions obtained from the 1

particle case. This works since we assume the particle positions to be similar to those solutions. To improve the coverage of our method, we consider both the positive solutions and mirror them to the negative axis. Mathematically, the starting guesses can be written as

$$\mathcal{Z}_i = \{z \quad | L_z(z) = 0\} \tag{3.26}$$

$$\mathcal{Z} = \mathcal{Z}_1 \otimes \mathcal{Z}_2 \otimes ... \otimes \mathcal{Z}_N \equiv \mathcal{Z}_i^N. \tag{3.27}$$

Alternatively, one may choose a very fine grid, which is very computationally heavy, since the number of starting points scales with the N-th power.

Using those results, we can begin to look at special values of N. In the next chapter, we will find the equilibrium positions of 2 particles. For the calculation, we used the same parameters in Table 2.1. Additionally, we kept $\kappa = 180\ 2\pi$kHz. All roots we compute are determined to be 0 in the error range of the numerical precision.

4. Considering the case of $N = 2$

When moving to higher dimensions, it gets difficult to visualize the solutions. The $N = 2$ case can still be visualized without getting too abstract.

For this special case, we choose a symmetric arrangement around the global origin where $d_1 = -d_2$. As such, we can describe the particle positions by $l := |d_1 - d_2| = 2|d_1|$. With that, we may reformulate our condition as

$$k_c \frac{l}{2} = n\pi. \tag{4.1}$$

This condition can be reformulated as

$$l = n\lambda_c. \tag{4.2}$$

To satisfy the assumption that the tweezers don't affect each other, we choose $n = 2$.

From the definition of the detuning (2.41), we can generate this expression

$$\lambda_c = \lambda_0 \frac{1}{\frac{\Delta}{\omega_0} + 1} \approx \lambda_0 \left(1 - \frac{\Delta}{\omega_0}\right) \tag{4.3}$$

which gives λ_c dependent on the detuning and thus l in dependence of the detuning. Since we only have two particles, we will define $\phi_1 = 0$ and $\phi_2 = \phi$.

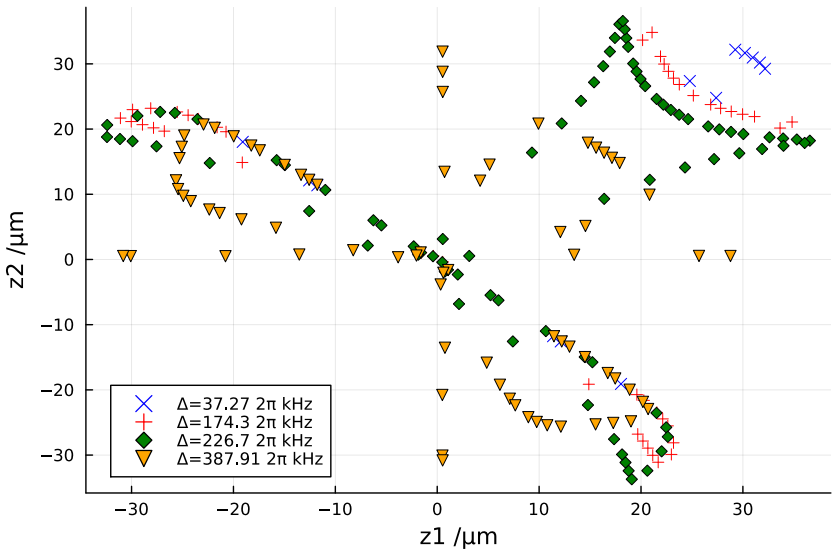
The results of the calculation can be seen in Figure 4.1a and Figure 4.1b. Figure 4.1a shows the system with a phase of $\phi = 0$ and Figure 4.1b shows the system with a phase of $\phi = \pi/10$. From Figure 4.1a we may see how the solutions lie in the quadrants I, II, and IV. [1] It can be observed that the solutions contract around the origin with rising detuning values. The solutions also lie along a loop in the II and IV quadrants, which become wider and move towards the origin with increasing detuning values.

It is also interesting to observe the asymmetry when adding a phase difference between the tweezers. As expected, the symmetry when exchanging z_1 and z_2 is broken when $\phi_1 \neq \phi_2$.

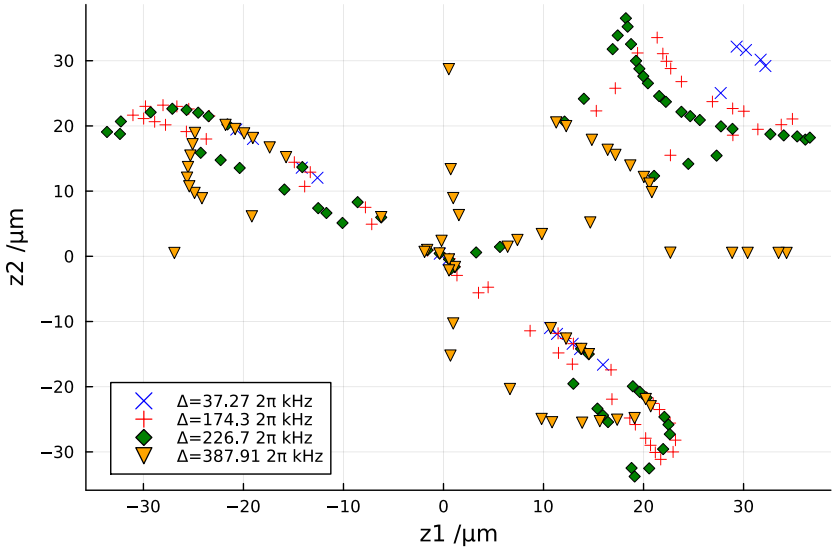
The solutions lie on a 1-dimensional manifold, when considering z_1 vs z_2. Since we can assume that the solutions vary smoothly with the detuning, we can deduce that the resulting solutions lie on a 2-dimensional manifold when plotting z_1 vs z_2 vs Δ. Note that the solutions lie on the manifold and do not constitute one.

We also computed the absolute value of the gradients around areas of this plot. From them, we can clearly see that there are distinct islands where the gradient vanishes, Which indicates that all solutions lie on a manifold, but not all points on the manifold are solutions. This also means that the equilibrium positions are discrete and they are not on a curve. The outputs of this calculation can be seen in Figure 4.2.

[1] Quadrant I...$z_1, z_2 > 0$ and II and IV...$z_1 = -z_2$



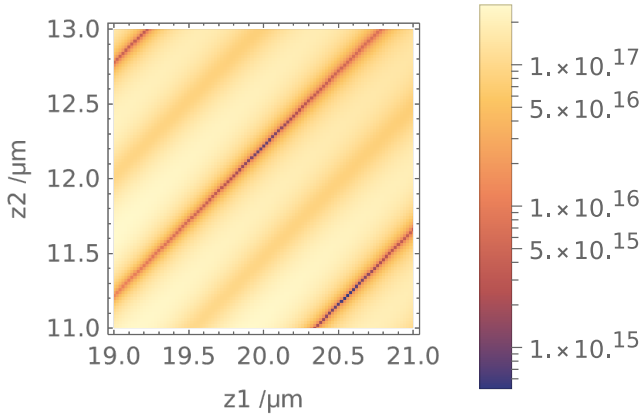
(a) Plot the solutions of $\phi = 0$.



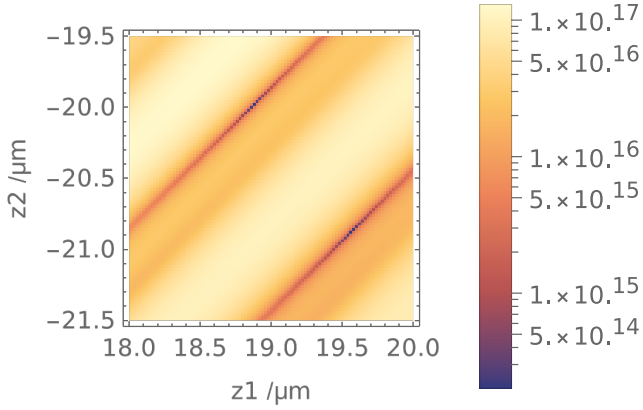
(b) Plot the solutions of $\phi = \pi/10$

Figure 4.1.: Plots of different ϕ values at different detuning values.

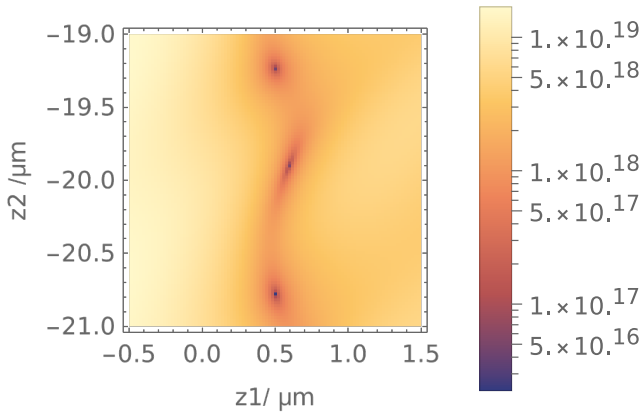
Additionally, $\kappa = 180\ 2\pi$kHz and the Parameters are from Table 2.1.



(a) Symmetric case: $z_1 \sim z_2$



(b) Antisymmetric Case: $z_1 \sim -z_2$



(c) Axes Case: $z_1 \sim 0$

Figure 4.2.: Heatmaps of $||\boldsymbol{L}||^2$

at $\Delta \simeq 387.91\ 2\pi\text{kHz}$

5. Summary

A theoretical framework was established to model N dielectric trapped particles in an optical cavity. This model provides a robust methodology for determining the equilibrium positions and characterizing the stability of those positions, specifically for the single particle case.

Also notable we discovered that with a single particle there are configurations with multiple stable equilibria. When examining the detuning vs the z displacement, we get a continuous curve, see Figure 2.3 and 2.2. Contrary to initial intuition, all equilibrium positions along this curve were found to be stable. This result challenges the expected bistable behavior, which only emerges when applying an adiabatic approximation (see Appendix B.2). This approximation fails, since the cavity needs sufficient time to reach its steady state. For 2 particles we found that for a fixed detuning the solutions are found as discrete points along a curve in configuration space. These results are also novel since it is typically assumed that κ is very large. Since we also consider smaller values of κ our predictions are very relevant for the field for further research, which is getting towards better cavities, where this domain can be explored.

This thesis establishes the fundamental principles required to transition this system into a quantum mechanical framework and explore emergent effects. Further research options are discussed extensively in Chapter 6.

6. Outlook and further Research Options

6.1. Quantum Effects

Even though the calculation done in this thesis was a purely classical one, the found equilibrium positions and fields are also valid when modeling the particles and fields as quantum degrees of freedom. For the single particle case, one can exploit the fact that there are multiple equilibrium positions that whose locations can be tuned via the tweezers detuning. For some particular detuning values they even merge. Thus, there are opportunities to potentially observe tunneling between these.

Increasing the number of particles is expected to yield rich physics coming from the interparticle interaction induced by the electromagnetic field [18], which will be amplified due to the presence of the cavity.

6.2. Generalizing the coordinates

In this thesis, we only looked at systems that satisfy $k_c y_i = n\pi$. It would be interesting to study the equilibrium positions when we relax this restriction and how this would affect the dynamics.

An interesting point to check would be if there exists a function $y_i(\boldsymbol{z})$ that keeps the system in an equilibrium. This would remove a degree of freedom for each particle and return the system to N conditions and N degrees of freedom.

In the $N = 2$ case, this would allow us to keep $l = const$ and vary Δ completely independently.

When doing those generalizations, it would also be interesting to try to find a potential that only depends on the particle positions directly. Currently, we took a look at $\boldsymbol{\nabla} V(\boldsymbol{z}, \alpha_c) \overset{!}{=} \boldsymbol{0}$ here we have α_c as an independent variable. And this degree of freedom is only implicitly constrained by the particle positions $\boldsymbol{z}$. It would provide meaningful insight if we could find a closed-form potential of $V = V(\boldsymbol{z})$. Since this would enable us to use the intuition that comes with a potential.

6.3. Higher Dimensions

The maximum number of particles we looked at was 2. For this amount, we may find that the solutions lie on a manifold, which has the same dimension as the number of particles. It would be interesting to look at how this changes for higher N and if this pattern continues.

It would also be interesting to see if there are configurations where no solution exists and the system cannot be brought into an equilibrium.

Another thing to work on for higher dimensions is to verify if the found positions are even stable. For the 1 particle case, we found that all positions are stable, but this may not be a pattern that continues ad infinitum. Finding an expression for N particles and checking the stability for higher N would yield very interesting insight.

6.4. Experimental verification

Of course, this work is purely theoretical up until this point. It would also be very interesting to see whether the predictions made here align with the experimental data and if any effects were neglected here that would need to be taken into account.

Thanks

My deepest gratitude goes to my supervisor, Carlos Gonzalez-Ballestero, who suggested this fascinating topic and whose guidance shaped much of the work presented here. His door was always open for questions and our regular meetings provided invaluable direction throughout the project.

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Bibliography

[1] C. Gonzalez-Ballestero et al. "Levitodynamics: Levitation and control of microscopic objects in vacuum". In: *Science* 374.6564 (2021), eabg3027. DOI: 10.1126/science.abg3027. eprint: https://www.science.org/doi/pdf/10.1126/science.abg3027. URL: https://www.science.org/doi/abs/10.1126/science.abg3027.

[2] A. Ashkin and J. M. Dziedzic. "Optical levitation in high vacuum". In: *Applied Physics Letters* 28.6 (Mar. 1976), pp. 333–335. ISSN: 0003-6951. DOI: 10.1063/1.88748. eprint: https://pubs.aip.org/aip/apl/article-pdf/28/6/333/18432654/333_1_online.pdf. URL: https://doi.org/10.1063/1.88748.

[3] A. Ashkin. "Acceleration and Trapping of Particles by Radiation Pressure". In: *Phys. Rev. Lett.* 24 (4 Jan. 1970), pp. 156–159. DOI: 10.1103/PhysRevLett.24.156. URL: https://link.aps.org/doi/10.1103/PhysRevLett.24.156.

[4] Dominik Windey et al. "Cavity-Based 3D Cooling of a Levitated Nanoparticle via Coherent Scattering". In: *Phys. Rev. Lett.* 122 (12 Mar. 2019), p. 123601. DOI: 10.1103/PhysRevLett.122.123601. URL: https://link.aps.org/doi/10.1103/PhysRevLett.122.123601.

[5] Stephan Troyer et al. *Quantum ground-state cooling of two librational modes of a nanorotor*. 2025. arXiv: 2509.13398 [quant-ph]. URL: https://arxiv.org/abs/2509.13398.

[6] Henning Rudolph et al. "Quantum Optical Binding of Nanoscale Particles". In: *Phys. Rev. Lett.* 133 (23 Dec. 2024), p. 233603. DOI: 10.1103/PhysRevLett.133.233603. URL: https://link.aps.org/doi/10.1103/PhysRevLett.133.233603.

[7] James Millen et al. "Optomechanics with levitated particles". In: *Reports on Progress in Physics* 83.2 (Jan. 2020), p. 026401. DOI: 10.1088/1361-6633/ab6100. URL: https://doi.org/10.1088/1361-6633/ab6100.

[8] David C Moore and Andrew A Geraci. "Searching for new physics using optically levitated sensors". In: *Quantum Science and Technology* 6.1 (Jan. 2021), p. 014008. DOI: 10.1088/2058-9565/abcf8a. URL: https://doi.org/10.1088/2058-9565/abcf8a.

[9] Nadav Priel et al. "Dipole moment background measurement and suppression for levitated charge sensors". In: *Science Advances* 8.41 (2022), eabo2361. DOI: 10.1126/sciadv.abo2361. eprint: https://www.science.org/doi/pdf/10.1126/sciadv.abo2361. URL: https://www.science.org/doi/abs/10.1126/sciadv.abo2361.

[10] Saurabh Gupta et al. *Quantum theory of electrically levitated nanoparticle-ion systems: Motional dynamics and sympathetic cooling.* 2025. arXiv: 2511.21495 [quant-ph]. URL: https://arxiv.org/abs/2511.21495.

[11] Patrick Maurer, Carlos Gonzalez-Ballestero, and Oriol Romero-Isart. "Quantum electrodynamics with a nonmoving dielectric sphere: quantizing Lorenz–Mie scattering". In: *Journal of the Optical Society of America B* 40.12 (Nov. 2023), p. 3137. ISSN: 1520-8540. DOI: 10.1364/josab.498540. URL: http://dx.doi.org/10.1364/JOSAB.498540.

[12] C. Gonzalez-Ballestero et al. "Theory for cavity cooling of levitated nanoparticles via coherent scattering: Master equation approach". In: *Physical Review A* 100.1 (July 2019). ISSN: 2469-9934. DOI: 10.1103/physreva.100.013805. URL: http://dx.doi.org/10.1103/PhysRevA.100.013805.

[13] Jeff Bezanson et al. "Julia: A fresh approach to numerical computing". In: *SIAM Review* 59.1 (2017), pp. 65–98. DOI: 10.1137/141000671. URL: https://epubs.siam.org/doi/10.1137/141000671.

[14] W. L. Erikson and Surendra Singh. "Polarization properties of Maxwell-Gaussian laser beams". In: *Phys. Rev. E* 49 (6 June 1994), pp. 5778–5786. DOI: 10.1103/PhysRevE.49.5778. URL: https://link.aps.org/doi/10.1103/PhysRevE.49.5778.

[15] John Verzani. *Roots.jl: Root finding functions for Julia.* https://github.com/JuliaMath/Roots.jl. 2020.

[16] J. Revels, M. Lubin, and T. Papamarkou. "Forward-Mode Automatic Differentiation in Julia". In: *arXiv:1607.07892 [cs.MS]* (2016). URL: https://arxiv.org/abs/1607.07892.

[17] J. E Dennis and Robert B Schnabel. *Numerical methods for unconstrained optimization and nonlinear equations*. eng. Prentice-Hall series in computational mathematics. Englewood Cliffs, N.J: Prentice Hall, 1983. ISBN: 0136272169.

[18] Jakob Rieser et al. "Tunable light-induced dipole-dipole interaction between optically levitated nanoparticles". In: *Science* 377.6609 (2022), pp. 987–990. DOI: 10.1126/science.abp9941. eprint: https://www.science.org/doi/pdf/10.1126/science.abp9941. URL: https://www.science.org/doi/abs/10.1126/science.abp9941.

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A. Notation

Name	**Used Symbol**
Natural Numbers	$\mathbb{N} = \{0, 1, 2, ...\}$
Integers	$\mathbb{Z}$
Rational Numbers	$\mathbb{Q}$
Real Numbers	$\mathbb{R}$
Complex Numbers	$\mathbb{C}$
Imaginary Unit	i
Complex conjugate	z^*
Real/Imaginary Part	$\Re z / \Im z$
Cartesian product	$\mathcal{M} \times \mathcal{N}$
Vectors	$\boldsymbol{v}$
Basis vectors	$\boldsymbol{e}_x, \boldsymbol{e}_y, \boldsymbol{e}_z$
Matrix	$\underline{\underline{A}}$
Transpose	$\underline{\underline{A}}^T$
Hermitian conjugate	$\underline{\underline{A}}^\dagger$
Diagonal Matrix	$\underline{\underline{Diag}}(v_1, ...v_n) = \begin{pmatrix} v_1 & & \\ & \ddots & \\ & & v_n \end{pmatrix}$
Scalar field	$f = f(\boldsymbol{v})$
Vector field	$\boldsymbol{F} = \boldsymbol{F}(\boldsymbol{v})$
Gradient	$\boldsymbol{\nabla} f$
Gradient w.r.t the variables $\boldsymbol{v} = [x, y, z]$	$\boldsymbol{\nabla}_{\boldsymbol{v}} f$
Gradient w.r.t the position of the n-th particle	$\boldsymbol{\nabla}_n f$
Divergence	$\boldsymbol{\nabla} \cdot \boldsymbol{F}$
Curl	$\boldsymbol{\nabla} \times \boldsymbol{F}$
Hessian	$\underline{\underline{\text{Hess}}} f$

Table A.1.: Table of Symbols

When applying a special set of variables, the operator will be annotated by a subscript.

B. Single Particle

B.1. Transformation of the Hessian

We have the Hamiltonian $H(\boldsymbol{r}, \boldsymbol{p}, a_c, a_c^*)$, we cannot calculate the Hessian directly with complex variables, since we can only take derivatives with respect to real variables. To get the Hessian, we reformulate the Hamiltonian to $H(\boldsymbol{r}, \boldsymbol{p}, \Re a, \Im a)$, which we then can take the Hessian of. In this formulation we define $\boldsymbol{\eta}' = [\boldsymbol{r}, \boldsymbol{p}, \Re a, \Im a]^T$. So the Hessian can be formulated as

$$\delta^2 H = \delta \boldsymbol{\eta}'^{\dagger} \cdot \underline{\underline{\text{Hess}}}_{\boldsymbol{\eta}'} H \cdot \delta \boldsymbol{\eta}'. \tag{B.1}$$

We then define a linear transform $\underline{\underline{T}}$, to satisfy $\boldsymbol{\eta} = \underline{\underline{T}} \cdot \boldsymbol{\eta}'$.

This fixes the transform $\underline{\underline{T}}$ to

$$\underline{\underline{T}} = \begin{pmatrix} \mathbb{1}_{dim\boldsymbol{r}+\dim \boldsymbol{p}} & & \\ & \frac{1}{2} & -\frac{\mathrm{i}}{2} \\ & \frac{1}{2} & \frac{\mathrm{i}}{2} \end{pmatrix}. \tag{B.2}$$

This transform restores the Wirtinger derivative

$$\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial \Re z} - \mathrm{i}\frac{\partial}{\partial \Im z}\right). \tag{B.3}$$

With those definitions we can rewrite the Equation B.1 to

$$\delta^2 H = \delta \boldsymbol{\eta}^{\dagger} \cdot \left(\underline{\underline{T}}^{-1}\right)^{\dagger} \cdot \underline{\underline{\text{Hess}}}_{\boldsymbol{\eta}'} H \cdot \underline{\underline{T}}^{-1} \cdot \delta \boldsymbol{\eta}, \tag{B.4}$$

which gives us the full Hessian for further analysis.

B.2. Keeping the Field at its equilibrium

To check the stability we view the System in a perturbed state $\boldsymbol{\eta} = \boldsymbol{\eta}_0 + \delta \boldsymbol{\eta}$. We do this for a small perturbation in z. Here we assume that a reaches its equilibrium much faster than z, such that $a = \alpha_c^{eq}(\boldsymbol{r})$ always holds. From Hamilton's equations we immediately get $\dot{z} = \partial_{p_z} H = p_z/m = 0$. With this, we know our dynamics are governed by $\ddot{z} = \dot{p}_z/m$, which we can get from the results in Equation (2.62).

$$\ddot{z} = \frac{\alpha}{m}|E_t|^2 L_z. \tag{B.5}$$

If $\ddot{z}$ points toward the equilibrium, we have a strong indicator that the System is stable and vice versa. A result of checking this stability can be seen in Figure B.1.

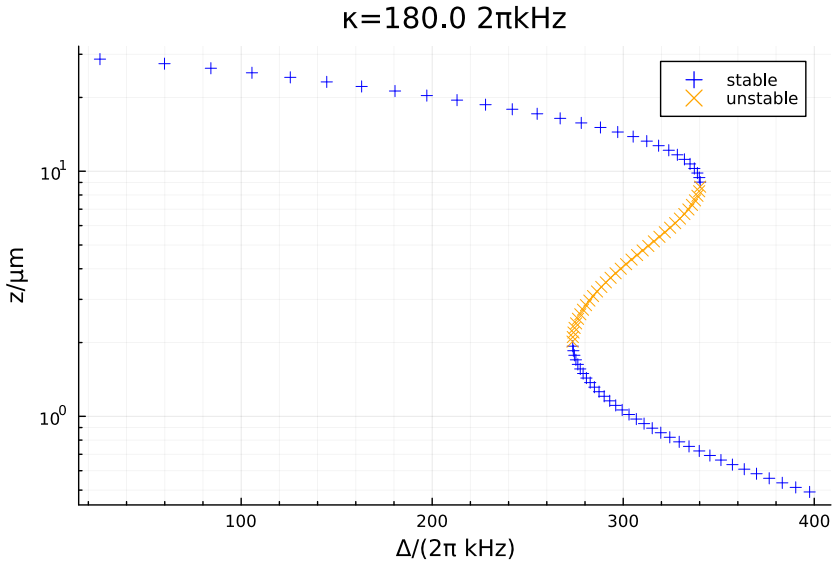

Figure B.1.: Plot of the equilibrium position at $\kappa = 180.0\ 2\pi kHz$ showing the stability of the equilibrium points with the parameters of Table 2.1

For this simulation, we assume that the system is stable if the restoring force points opposite to the perturbation.

Here we get the optical bistability we would expect from intuition. However, the approximation is not physical, since the cavity needs a finite non-zero relaxation time, that will not be instantaneous.